4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- **Note that $\tau_{xy,web}$ >> $\tau_{xy,flange}$; in fact, *min*($\tau_{xy,web}$) is greater than *max*($\tau_{xy,flange}$) by a factor of 12**

- **This is the usual trend with thin-walled beams and in general, the thinner the member, the greater the disparity**

- **For any segment of a cross section where one dimension is much "thinner" than the other, it is common practice to neglect the shear stress pointing parallel to the "thinner" dimension**

- **For this reason, we will consider $\tau_{xz,web}$ and $\tau_{xy,flange}$ in this cross section to be negligible, while $\tau_{xy,web}$ and $\tau_{xz,flange}$ are non-negligible**

4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- **Let us then use $\tau_{xz} = \frac{VQ}{Ib}$ to find τ_{xz} as a function of *z* in each flange:**

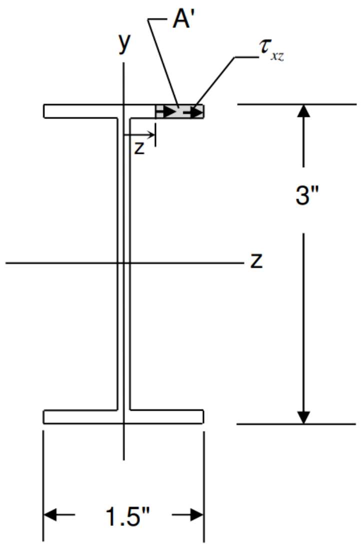
- ***V* and *I* are the same values as before, *b = (1/8)in*, and $Q = \bar{y}'A'$ where *z* varies from *0* to *0.75in***

- **In this case $\bar{y}'$ is constant at *(1.5 – 1/16)in*, while A' varies with *z***

- **Result is**

$$\tau_{xz} = \frac{(1{,}400\ lb)\left[\left(1.5"-\frac{1}{16}"\right)(0.75"-z)\left(\frac{1}{8}"\right)\right]}{\left(0.9920\ in^4\right)\left(\frac{1}{8}"\right)}$$

$$\boxed{\left(\tau_{xz}\right)_{\text{Flanges}} = (2{,}029)\left[(0.75)-z\right]}$$



Thicknesses: 1/8"

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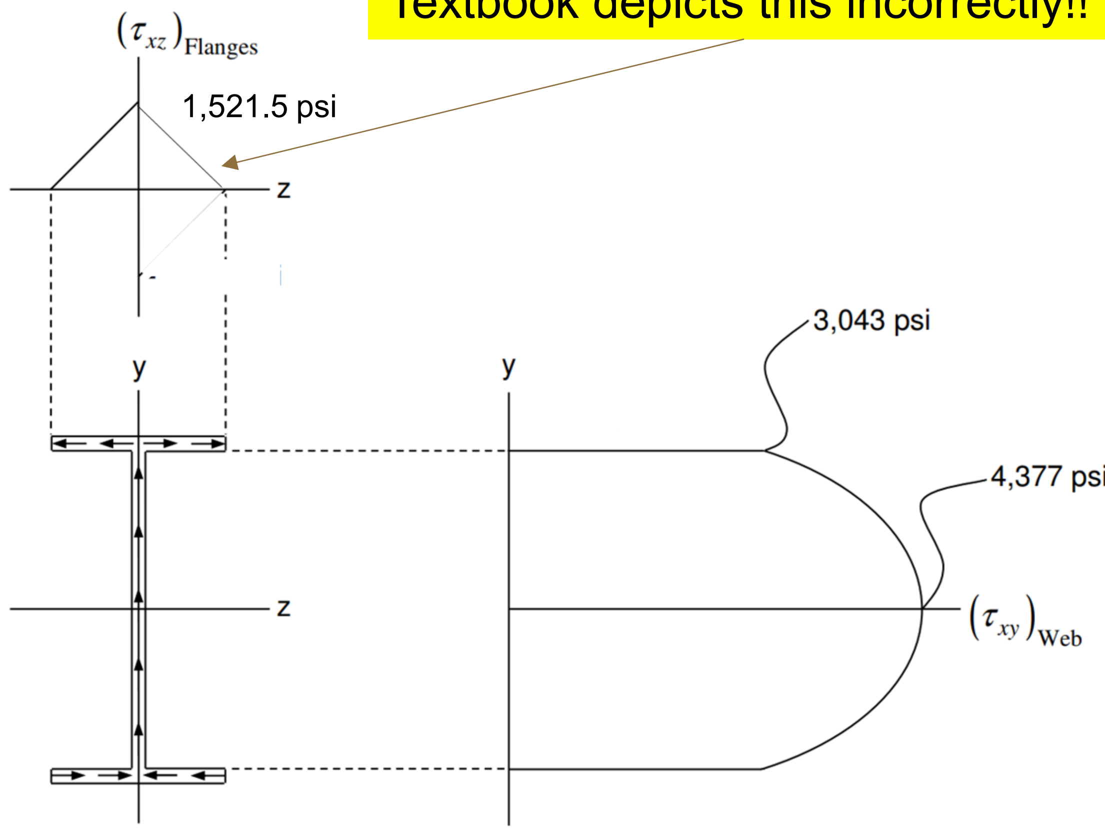
4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- **For the upper flange, when** $z > 0$**,** τ_{xz} **varies from** *1521.5 psi* **at the midpoint of the flange** *(z = 0)* **to** *0* **at the right edge** *(z = 0.75in)*

- **For the upper flange, when** $z < 0$**,** τ_{xz} **is a mirror image of the above**

- **We can now plot** τ **variation as a function of** *y* **in the web and** *z* **in the flanges**

Textbook depicts this incorrectly!!



Note that $\tau_{xz,flange}$ is greater than $\tau_{xy,flange}$ by a factor of 6 and that $\tau_{xz,flange}$ is not negligible with respect to $\tau_{xy,web}$ → $min(\tau_{xy,web}) = 2*max(\tau_{xz,flange})$

4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- **Here we introduce the concept of shear flow, q, in a structure**

- **It is advantageous for thin-walled beams and defined by $q = \tau \cdot b$ where b is the thickness at that point in the structure**

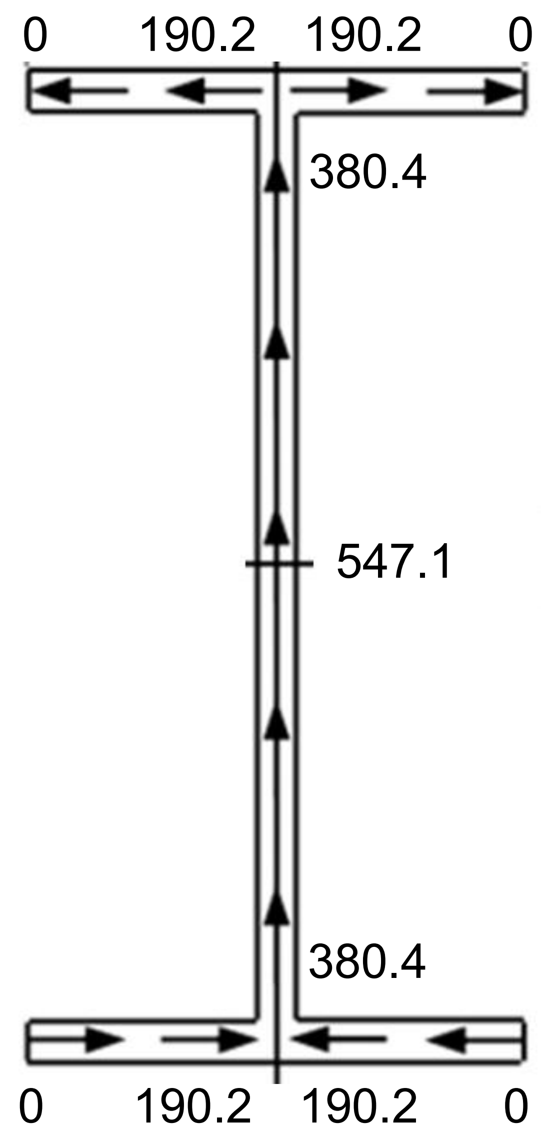
- **Typical units are $\frac{lb}{in}$**

- **Inserting the familiar shear stress formula $\tau = \frac{VQ}{Ib}$ yields $q = \frac{VQ}{I}$**

- **Since the thickness of the I-beam in the previous example is constant at 1/8", the τ distribution can be converted to q by simply multiplying by 1/8**
  - Maintains the parabolic distribution (or "flow") in the web and linear distribution in the flanges

4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- **Let's examine the value of** q **at various points along this I-beam**
- **Beginning at the lower left and lower right corners,** $q = 0$ **(due to the free edges)**
- **Then at the junction between the lower flange and the web,** q **increases to** *(1521.5/8) = 190.2 lb/in*
- **Going up the web,** q **increases from** *(3043/8) = 380.4 lb/in* **to a max of** *(4377/8) = 547.1 lb/in* **and back to** *380.4*
- **Finally decreases from** *190.2* **to** *0* **to the left and to the right**



4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- **Since** *380.4 = 2*190.2*, **this behavior is very analogous to fluid flow!**

- **Since thickness is constant everywhere, q and τ are proportional, so the fluid analogy holds for τ as well**

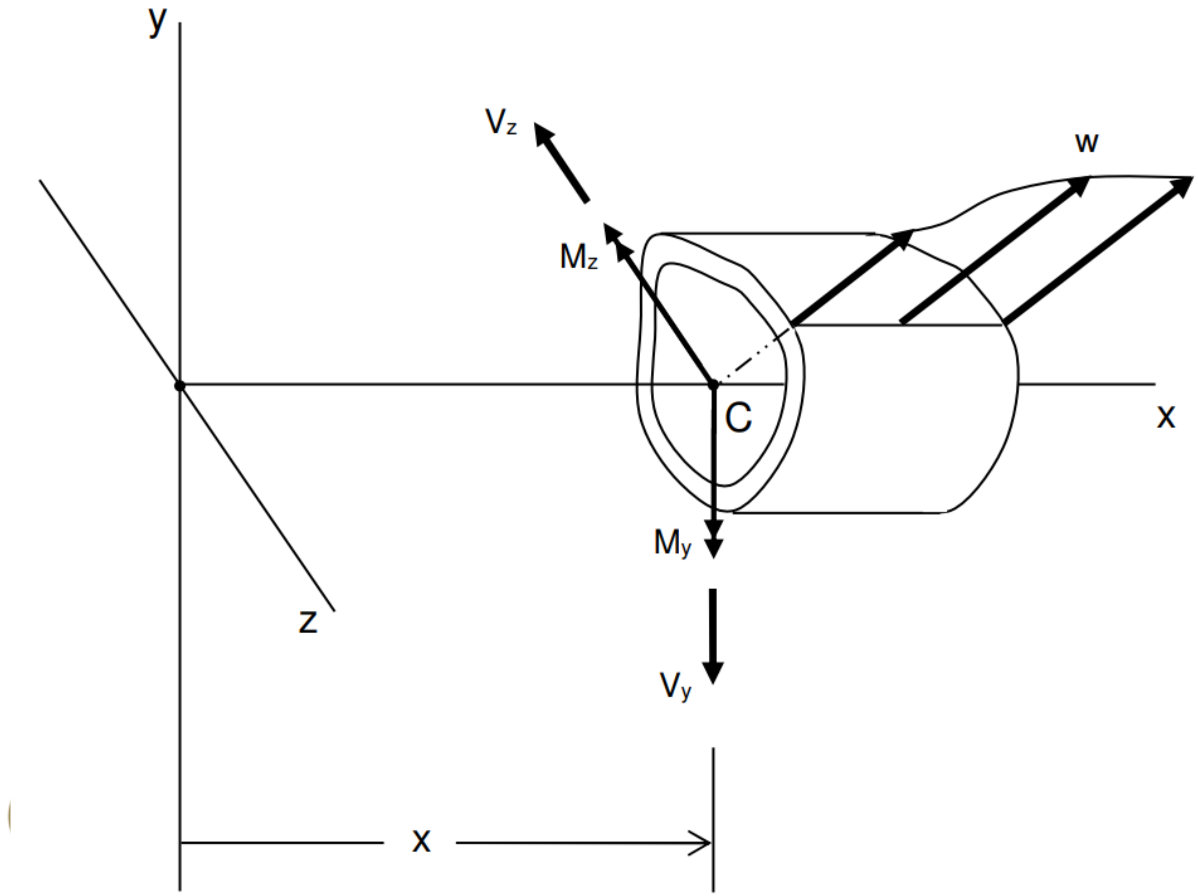
- **This is generally true for any scenario involving a beam of constant thickness; when thickness is not constant, the analogy no longer holds for τ, but it still holds for q!**

- **Just as we learned the equation $\sigma = \frac{My}{I}$ only holds under certain assumptions, the equation $\tau = \frac{VQ}{Ib}$ (and therefore $q = \frac{VQ}{I}$) is also a special case**

- **Here we derive the General Shear Flow Formula --------------------→**

4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

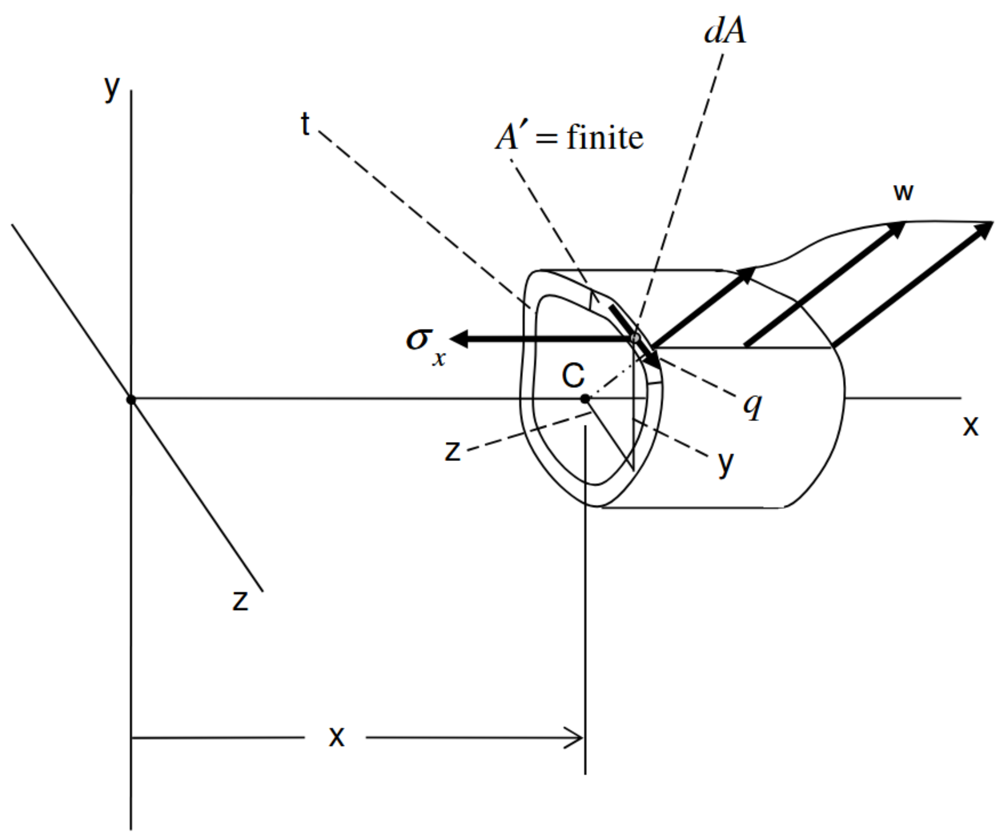
- **Consider the thin-walled beam section below, where the only applied load is a general transverse force (here depicted as a distributed load) and the corresponding internal reactions (i.e. *y* and *z* forces and moments) are shown:**



4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- Here is another way to represent the beam section, with a normal stress distribution resulting from the internal bending moment and shear stress distribution (or, alternately, shear flow) resulting from the internal shear force

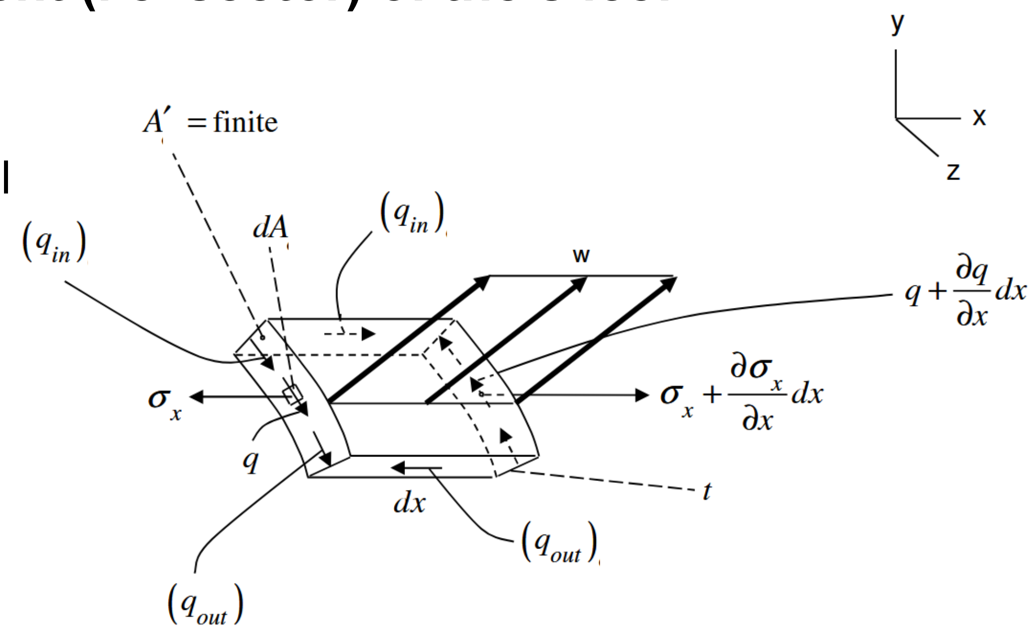
- Notice that only the shear flow perpendicular to the thickness has been retained (shear flow parallel to the thickness is assumed negligible)



4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- **Suppose we cut a "slice" of the section of infinitesimal length *dx* and consider a certain element (i.e. sector) of the slice:**

- **Note that:**
  - The element is only infinitesimal in the *x* direction
  - The two *x* faces experience normal stress and shear stress (depicted as shear flow)
  - The faces comprising the inner and outer wall experience no shear stress because they are "free" (although the distributed load is present)
  - The remaining two faces experience shear flow labeled "q_{in}" and "q_{out}"



In this sketch and the following equations, the textbook chooses to employ a "σ" subscript. It is confusing → ignore it!

4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- **Let's assume that *q* is constant across the "*dx*" thickness, but can vary along the circumference of the cross-section**

- **Therefore q_{in} and q_{out} are the *q* "flowing" into and out of the area *A'*, respectively**

- **The only forces acting in the x direction are those due to q_{in}, q_{out}, σ_x, and $\sigma_x + \frac{d\sigma_x}{dx}dx$**

4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- **Summing forces in the *x* direction and setting them = 0 yields:**

$$-\left(q_{out}\right)\ dx+\left(q_{in}\right)\ dx-\int_{A'}\sigma_x\,dA+\int_{A'}\left(\sigma_x+\frac{\partial\sigma_x}{\partial x}dx\right)dA=0$$

$$\Rightarrow\qquad\left[\left(q_{out}\right)\ -\left(q_{in}\right)\ \right]dx=\int_{A'}\left(\frac{\partial\sigma_x}{\partial x}dx\right)dA$$

$$\boxed{\left(q_{out}\right)\ -\left(q_{in}\right)\ =\int_{A'}\frac{\partial\sigma_x}{\partial x}\,dA}$$

4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- **Recall the General Flexure Formula for σ_x in the cross-section, valid in the case of pure bending:**

$$\sigma_x = \frac{-y\left(I_y M_z + I_{yz} M_y\right) + z\left(I_{yz} M_z + I_z M_y\right)}{I_y I_z - {I_{yz}}^2}$$

- **In the current situation, the loading is not pure bending (i.e. applied moment), but transverse loading (i.e. applied force)**

- **However, if the beam is a long, slender beam (*L* >> largest cross-sectional dimension), then using the General Flexure Formula is a good approximation to evaluate σ_x**

- **Another caveat for using this formula is that the origin of the *yz* frame lies along the neutral axis (typically at the centroid); we will assume this is the case here**